T3 E_+0: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

E_+0

Additional quartics: 6, additional bilinears: 1, extra Yukawa: y1Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W^A_\mu$
g_3	$D_\mu \supset ig_3 t^A G^A_\mu$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$LFS_1 + \text{h.c.}$
y_2	$LFS_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 HHS_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) HH$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\mathrm{BSM}} =$ `Coupling(lambdaT3ExtraM000200I1, {}, 0)CG(T3ExtraCGM000200I1, {Index(d$$1, T3BSMd3), Index`

`+Coupling(lambdaT3ExtraQ000400I1, {}, 0)Bar(Field(NewScalar1, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar1, Scalar, {Index(d$$2, T3BSMd3)}, {}))Bar(Field(NewScalar1, Scalar, {Index(d$$3, T3BSMd3)}, {}))Bar(Field(NewScalar1, Scalar, {Index(d$$4, T3BSMd3)}, {}))CG(T3ExtraCGQ000400I1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$4, T3BSMd3)})`

$$+\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ110200I1},\{\},0)\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ110200I1},\{\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Bar}(\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ021010I1},\{\},0))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{CG}(\mathrm{Bar}(\mathrm{T3ExtraCGQ021010I1}),\{\mathrm{Bar}(\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{MS1},\{\},0)^{2}\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})(-\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})))$$

$$+\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\mathrm{Index}(d\$\$1,\mathrm{Lorentz})\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\mathrm{Index}(d\$\$1,\mathrm{Lorentz})\}))$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaH1Inv1},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3HiggsPortal1CGInv1},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaH1Inv1},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3HiggsPortal1CGInv1},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaH1Inv2},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3HiggsPortal1CGInv2},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaH1Inv2},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3HiggsPortal1CGInv2},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{lambdaT3},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{CG}(\mathrm{T3MixCG},\{\mathrm{Bar}(\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Bar}(\mathrm{Coupling}(\mathrm{lambdaT3ExtraM000200I1},\{\},0))\mathrm{CG}(\mathrm{T3ExtraCGM000200I1},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})$$

$$+\mathrm{Bar}(\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ110200I1},\{\},0))\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ110200I1},\{\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ001300I1},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ001300I1},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaS1Inv1},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3Scalar1SelfCGInv1},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaS1Inv2},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3Scalar1SelfCGInv2},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Bar}(\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ001300I1},\{\},0))\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ001300I1},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Bar}(\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ000400I1},\{\},0))\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{CG}(\mathrm{T3ExtraCGQ000400I1},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{MS2},\{\},0)^{2}\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})(-\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})))$$

$$+\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\mathrm{Index}(d\$\$1,\mathrm{Lorentz})\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\mathrm{Index}(d\$\$1,\mathrm{Lorentz})\}))$$

$$+\mathrm{Bar}(\mathrm{Coupling}(\mathrm{lambdaT3},\{\},0))\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{Bar}(\mathrm{T3MixCG}),\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaH2Inv1},\{\},0)\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3HiggsPortal2CGInv1},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaH2Inv1},\{\},0)\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3HiggsPortal2CGInv1},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaH2Inv2},\{\},0)\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3HiggsPortal2CGInv2},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaH2Inv2},\{\},0)\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3HiggsPortal2CGInv2},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ021010I1},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{Bar}(\mathrm{Field}(H,\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund}))\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ021010I1},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$2,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ000211I1},\{\},0)\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ000211I1},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ000211I2},\{\},0)\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ000211I2},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{lambda12Inv1},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3CrossScalarCGInv1},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{lambda12Inv2},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3CrossScalarCGInv2},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Coupling}(\mathrm{lambda12Inv3},\{\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3CrossScalarCGInv3},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Bar}(\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ000211I1},\{\},0))\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ000211I1},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\mathrm{Bar}(\mathrm{Coupling}(\mathrm{lambdaT3ExtraQ000211I2},\{\},0))\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3ExtraCGQ000211I2},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaS2Inv1},\{\},0)\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3Scalar2SelfCGInv1},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$+\frac{1}{2}\mathrm{Coupling}(\mathrm{lambdaS2Inv2},\{\},0)\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3})\},\{\})\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3Scalar2SelfCGInv2},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$3,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$2,\mathrm{T3BSMd3}),\mathrm{Index}(d\$\$4,\mathrm{T3BSMd3})\})$$

$$-\mathrm{Coupling}(\mathrm{MF},\{\},0)\mathrm{NCM}(\mathrm{Bar}(\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\})),\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\}))$$

$$+\mathrm{Coupling}(y2,\{\mathrm{Index}(d\$\$1,\mathrm{Flavor}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},0)\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3Y2CG},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Bar}(\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})\mathrm{NCM}(\mathrm{Bar}(\mathrm{Field}(l,\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{Flavor})\},\{\})),\mathrm{DiracProduct}(\mathrm{Proj}(1)),\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\}))$$

$$+\mathrm{Coupling}(\mathrm{y1Conj},\{\mathrm{Index}(d\$\$1,\mathrm{Flavor}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},0)\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{T3Y1ConjCG},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})\mathrm{NCM}(\mathrm{Bar}(\mathrm{Field}(l,\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{Flavor})\},\{\})),\mathrm{DiracProduct}(\mathrm{GammaCC},\mathrm{Proj}(1)),\mathrm{Transp}(\mathrm{Bar}(\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\}))))$$

$$+\mathrm{Coupling}(y1,\{\mathrm{Index}(d\$\$1,\mathrm{Flavor}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},0)\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})\mathrm{CG}(\mathrm{T3Y1CG},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})\mathrm{NCM}(\mathrm{Bar}(\mathrm{Field}(l,\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{Flavor})\},\{\})),\mathrm{DiracProduct}(\mathrm{GammaCC},\mathrm{Proj}(1)),\mathrm{Transp}(\mathrm{Bar}(\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\}))))$$

$$+i\mathrm{NCM}(\mathrm{Bar}(\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\})),\mathrm{DiracProduct}(\mathrm{GammaM}(\mathrm{Index}(d\$\$1,\mathrm{Lorentz}))),\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\mathrm{Index}(d\$\$1,\mathrm{Lorentz})\}))$$

$$+\mathrm{Field}(\mathrm{NewScalar2},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Coupling}(y2,\{\mathrm{Index}(d\$\$1,\mathrm{Flavor}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},0))\mathrm{CG}(\mathrm{Bar}(\mathrm{T3Y2CG}),\{\mathrm{Bar}(\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})\mathrm{NCM}(\mathrm{Bar}(\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\})),\mathrm{DiracProduct}(\mathrm{Proj}(-1)),\mathrm{Field}(l,\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{Flavor})\},\{\}))$$

$$+\mathrm{Bar}(\mathrm{Coupling}(y1,\{\mathrm{Index}(d\$\$1,\mathrm{Flavor}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},0))\mathrm{Bar}(\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\}))\mathrm{CG}(\mathrm{Bar}(\mathrm{T3Y1CG}),\{\mathrm{Bar}(\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Bar}(\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})\mathrm{NCM}(\mathrm{Transp}(\mathrm{Field}(l,\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{Flavor})\},\{\})),\mathrm{DiracProduct}(\mathrm{GammaCC},\mathrm{Proj}(-1)),\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\}))$$

$$+\mathrm{Field}(\mathrm{NewScalar1},\mathrm{Scalar},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\},\{\})\mathrm{Bar}(\mathrm{Coupling}(\mathrm{y1Conj},\{\mathrm{Index}(d\$\$1,\mathrm{Flavor}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},0))\mathrm{CG}(\mathrm{Bar}(\mathrm{T3Y1ConjCG}),\{\mathrm{Bar}(\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund}))),\mathrm{Bar}(\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2})),\mathrm{Index}(d\$\$1,\mathrm{T3BSMd3})\})\mathrm{NCM}(\mathrm{Transp}(\mathrm{Field}(l,\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{SU2L}(\mathrm{fund})),\mathrm{Index}(d\$\$1,\mathrm{Flavor})\},\{\})),\mathrm{DiracProduct}(\mathrm{GammaCC},\mathrm{Proj}(-1)),\mathrm{Field}(\mathrm{NewFermion},\mathrm{Fermion},\{\mathrm{Index}(d\$\$1,\mathrm{T3BSMd2}),\mathrm{Index}(d\$\$1,\mathrm{NFlavor})\},\{\}))$$

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Yukawa1ConjugateScalar_GenericCG`
- `Scalar1SelfInv1`

- `Scalar1SelfInv2`
- `Scalar2SelfInv1`
- `Scalar2SelfInv2`
- `HiggsPortal1Inv1`
- `HiggsPortal1Inv2`
- `HiggsPortal2Inv1`
- `HiggsPortal2Inv2`
- `ScalarCrossPortalInv1`
- `ScalarCrossPortalInv2`
- `ScalarCrossPortalInv3`
- `T3ScalarMix_GenericCG`
- `T3ExtraQ_0_0_0_2_1_1_1`
- `T3ExtraQ_0_0_0_2_1_1_2`
- `T3ExtraQ_0_0_0_4_0_0_1`
- `T3ExtraQ_0_0_1_3_0_0_1`
- `T3ExtraQ_0_2_1_0_1_0_1`
- `T3ExtraQ_1_1_0_2_0_0_1`
- `T3ExtraM_0_0_0_2_0_0_1`

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2\,\frac{dg_1}{d\ln\mu} = \frac{59{g_1}^3}{6}$$

g_2

$$16\pi^2\,\frac{dg_2}{d\ln\mu} = \frac{{g_2}^3}{6}$$

g_3

$$16\pi^2\,\frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}16\pi^2\,\frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij}\\ & + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij}\\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij}\\ & - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij})\\ & - \frac{9}{4}{g_2}^2\,(Y_u)_{ij}\\ & - 8{g_3}^2\,(Y_u)_{ij}\\ & - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}\end{aligned}$$

Y_d

$$\begin{aligned}16\pi^2\,\frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}\\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}\\ & + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}\\ & + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij})\\ & - \frac{9}{4}{g_2}^2\,(Y_d)_{ij}\\ & - 8{g_3}^2\,(Y_d)_{ij}\\ & - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}\end{aligned}$$

Y_e

$$
\begin{aligned}
16\pi^2 \frac{dY_e}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\
& + \frac{3}{4}(\Big(y_1\ (y_1)^\dagger\ Y_e\Big)_{ij} + \Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ Y_e\Big)_{ij} + \Big(y_2\ (y_2)^\dagger\ Y_e\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\
& - \frac{15}{4}{g_1}^2\,(Y_e)_{ij}
\end{aligned}
$$

y_1

$$
\begin{aligned}
16\pi^2 \frac{dy_1}{d\ln\mu} = {} & \mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big)\,(y_1)_{ij} \\
& + 2\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big)\,(\mathrm{y1Conj})_{ij} \\
& + T_\nu^{(1)}\,(y_1)_{ij} \\
& + \frac{1}{12}(29\,\Big(y_1\ (\mathrm{y1Conj})^\dagger\ \mathrm{y1Conj}\Big)_{ij} + 29\,\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ y_1\Big)_{ij} + 9\,\Big(y_2\ (y_2)^\dagger\ y_1\Big)_{ij} + 6\,\Big(Y_e\ (Y_e)^\dagger\ y_1\Big)_{ij} + 18\,\Big(\\
& - \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(y_1)_{ij}
\end{aligned}
$$

y_2

$$
\begin{aligned}
16\pi^2 \frac{dy_2}{d\ln\mu} = {} & T_\nu^{(2)}\,(y_2)_{ij} \\
& + \frac{1}{4}(3\,\Big(y_1\ (y_1)^\dagger\ y_2\Big)_{ij} + 3\,\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ y_2\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ y_2\Big)_{ij} + 6\,\Big(y_2\ (y_2)^\dagger\ y_2\Big)_{ij}) \\
& - \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(y_2)_{ij}
\end{aligned}
$$

M_F

$$
\begin{aligned}
16\pi^2 \frac{dM_F}{d\ln\mu} = {} & \frac{3}{4}(\Big((y_1)^T\ (y_1)^*\ M_F\Big)_{ij} + \Big((\mathrm{y1Conj})^T\ (\mathrm{y1Conj})^*\ M_F\Big)_{ij} + \Big(M_F\ (y_2)^\dagger\ y_2\Big)_{ij}) \\
& - \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\
& - \frac{3}{2}{g_1}^2\,(M_F)_{ij}
\end{aligned}
$$

m_1^2

$$
16\pi^2 \frac{dm_1^2}{d\ln\mu} = m_1^2(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})
$$

$$\begin{aligned}
&+ \mathrm{muT3GeneralS1Sq}(4\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^{\dagger}\Big) + 20\,(\mathrm{lambdaT3GeneralQ001300I1})^{*})\\
&- 4\,\mathrm{Tr}\Big(M_F\ (M_F)^{\dagger}\ (y_1)^{T}\ (y_1)^{*}\Big)\\
&- 4\,\mathrm{Tr}\Big(M_F\ (M_F)^{\dagger}\ (\mathrm{y1Conj})^{T}\ (\mathrm{y1Conj})^{*}\Big)\\
&+ 4\,(\mathrm{muT3GeneralS1Sq})^{*}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^{\dagger}\Big)\\
&+ 20\mathrm{lambdaT3GeneralQ001300I1}\,(\mathrm{muT3GeneralS1Sq})^{*}\\
&- 12{g_2}^2 m_1^2\\
&+ m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})
\end{aligned}$$

m_2^2

$$\begin{aligned}
16\pi^2\,\frac{dm_2^2}{d\ln\mu} &= m_2^2(2T_{\nu}^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})\\
&- 4\,\mathrm{Tr}\Big(M_F\ (y_2)^{\dagger}\ y_2\ (M_F)^{\dagger}\Big)\\
&+ 4\mathrm{muT3GeneralS1Sq}(3\,(\mathrm{lambdaT3GeneralQ000211I1})^{*} + (\mathrm{lambdaT3GeneralQ000211I2})^{*})\\
&+ 12\mathrm{lambdaT3GeneralQ000211I1}\,(\mathrm{muT3GeneralS1Sq})^{*}\\
&+ 4\mathrm{lambdaT3GeneralQ000211I2}\,(\mathrm{muT3GeneralS1Sq})^{*}\\
&- 12{g_2}^2 m_2^2\\
&- 6{g_1}^2 m_2^2\\
&+ m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})
\end{aligned}$$

λ_1

$$\begin{aligned}
16\pi^2\,\frac{d\lambda_1}{d\ln\mu} &= \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^{\dagger}\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^{\dagger}\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^{\dagger}\Big))\\
&- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^{\dagger}\ Y_d\ (Y_d)^{\dagger}\Big)\\
&- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\Big)\\
&- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^{\dagger}\ Y_u\ (Y_u)^{\dagger}\Big)\\
&+ 4\lambda_5\,(\lambda_5)^{*}\\
&+ 4\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^{*}\\
&+ 24\mathrm{lambdaT3GeneralQ110200I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^{*}\\
&+ \frac{9{g_2}^4}{4}\\
&+ {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right)\\
&- 9{g_2}^2\lambda_1\\
&+ \frac{3{g_1}^4}{4}\\
&+ 12{\lambda_1}^2\\
&+ 6{\lambda_{HS_1}^{(1)}}^2\\
&+ 3{\lambda_{HS_1}^{(A)}}^2\\
&+ 6{\lambda_{HS_2}^{(1)}}^2\\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}$$

$\lambda^{(1)}_{S_1}$

$$\begin{aligned}
16\pi^2\,\frac{d\lambda^{(1)}_{S_1}}{d\ln\mu} &= \lambda^{(1)}_{S_1}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^{\dagger}\Big) + 8\lambda^{(A)}_{S_1}) \\
&+ \mathrm{lambdaT3GeneralQ001300I1}\left(\frac{32}{3}\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^{\dagger}\Big) + \frac{352\,(\mathrm{lambdaT3GeneralQ001300I1})^{*}}{3}\right) \\
&+ \frac{32}{3}\,(\mathrm{lambdaT3GeneralQ001300I1})^{*}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^{\dagger}\Big) \\
&- \frac{248}{45}\,\mathrm{Tr}\Big(y_1\ (y_1)^{\dagger}\ \mathrm{y1Conj}\ (\mathrm{y1Conj})^{\dagger}\Big) \\
&- \frac{248}{45}\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^{\dagger}\ \mathrm{y1Conj}\ (y_1)^{\dagger}\Big) \\
&- \frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_1\ (y_1)^{\dagger}\Big) \\
&- \frac{8}{3}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^{\dagger}\ \mathrm{y1Conj}\ (\mathrm{y1Conj})^{\dagger}\Big) \\
&+ \frac{8}{3}\mathrm{lambdaT3GeneralQ000211I1}(3\,(\mathrm{lambdaT3GeneralQ000211I1})^{*} + (\mathrm{lambdaT3GeneralQ000211I2})^{*}) \\
&+ \mathrm{lambdaT3GeneralQ000211I2}\left(\frac{8\,(\mathrm{lambdaT3GeneralQ000211I1})^{*}}{3} + 4\,(\mathrm{lambdaT3GeneralQ000211I2})^{*}\right) \\
&+ \frac{832}{3}\mathrm{lambdaT3GeneralQ000400I1}\,(\mathrm{lambdaT3GeneralQ000400I1})^{*} \\
&+ \frac{16}{3}\mathrm{lambdaT3GeneralQ110200I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^{*} \\
&+ 22{g_2}^{4} \\
&- 24{g_2}^{2}\lambda^{(1)}_{S_1} \\
&+ 6{\lambda^{(1)}_{12}}^{2} \\
&+ 8\lambda^{(1)}_{12}\lambda^{(\times)}_{12} \\
&+ \frac{20{\lambda^{(\times)}_{12}}^{2}}{3} \\
&+ 4{\lambda^{(1)}_{HS_1}}^{2} \\
&+ 14{\lambda^{(1)}_{S_1}}^{2} \\
&+ 16{\lambda^{(A)}_{S_1}}^{2}
\end{aligned}$$

$\lambda^{(1)}_{S_2}$

$$\begin{aligned}
16\pi^2\,\frac{d\lambda^{(1)}_{S_2}}{d\ln\mu} &= \lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 8\lambda^{(A)}_{S_2}) \\
&- \frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big) \\
&+ 8\mathrm{lambdaT3GeneralQ000211I1}(3\,(\mathrm{lambdaT3GeneralQ000211I1})^{*} + (\mathrm{lambdaT3GeneralQ000211I2})^{*}) \\
&+ \mathrm{lambdaT3GeneralQ000211I2}\left(8\,(\mathrm{lambdaT3GeneralQ000211I1})^{*} + \frac{20\,(\mathrm{lambdaT3GeneralQ000211I2})^{*}}{3}\right) \\
&+ 22{g_2}^{4} \\
&+ {g_1}^{2}\left(8{g_2}^{2} - 12\lambda^{(1)}_{S_2}\right) \\
&- 24{g_2}^{2}\lambda^{(1)}_{S_2} \\
&+ 12{g_1}^{4} \\
&+ 6{\lambda^{(1)}_{12}}^{2}
\end{aligned}$$

$$+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$$
$$+ \frac{20{\lambda_{12}^{(\times)}}^2}{3}$$
$$+ 4{\lambda_{HS_2}^{(1)}}^2$$
$$+ 14{\lambda_{S_2}^{(1)}}^2$$
$$+ 16{\lambda_{S_2}^{(A)}}^2$$

$\lambda_{HS_1}^{(1)}$

$$16\pi^2\,\frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = \lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_1}^{(A)})$$
$$+ \mathrm{lambdaT3GeneralQ110200I1}(4\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big) + 4(5\,(\mathrm{lambdaT3GeneralQ001300I1})^* + 4\,(\mathrm{lambdaT3Gene}$$
$$+ 4\,(\mathrm{lambdaT3GeneralQ110200I1})^*\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^\dagger\Big)$$
$$- 2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$$
$$- 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ Y_e\ (Y_e)^\dagger\Big)$$
$$+ 4\lambda_5\,(\lambda_5)^*$$
$$+ 20\mathrm{lambdaT3GeneralQ001300I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^*$$
$$+ 4\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^*$$
$$+ 6{g_2}^4$$
$$- \frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$$
$$- \frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$$
$$+ \lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$$
$$+ 6\lambda_1\lambda_{HS_1}^{(1)}$$
$$+ 4{\lambda_{HS_1}^{(1)}}^2$$
$$+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$$
$$+ 6{\lambda_{HS_1}^{(A)}}^2$$

$\lambda_{HS_2}^{(1)}$

$$16\pi^2\,\frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = \lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)})$$
$$- 2\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$$
$$+ 4\lambda_5\,(\lambda_5)^*$$
$$+ 4\mathrm{lambdaT3GeneralQ110200I1}(3\,(\mathrm{lambdaT3GeneralQ000211I1})^* + (\mathrm{lambdaT3GeneralQ000211I2})^*)$$
$$+ 12\mathrm{lambdaT3GeneralQ000211I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^*$$
$$+ 4\mathrm{lambdaT3GeneralQ000211I2}\,(\mathrm{lambdaT3GeneralQ110200I1})^*$$
$$+ 4\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^*$$
$$+ 6{g_2}^4$$
$$- \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$$

$$
\begin{aligned}
&+ 3{g_1}^4 \\
&- \frac{15{g_1}^2\lambda^{(1)}_{HS_2}}{2} \\
&+ \lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}) \\
&+ 6\lambda_1\lambda^{(1)}_{HS_2} \\
&+ 4{\lambda^{(1)}_{HS_2}}^2 \\
&+ 8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2} \\
&+ 6{\lambda^{(A)}_{HS_2}}^2
\end{aligned}
$$

$\lambda^{(1)}_{12}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda^{(1)}_{12}}{d\ln\mu} = {}& \lambda^{(1)}_{12}(2T^{(1)}_\nu + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 2T^{(2)}_\nu + 4\lambda^{(A)}_{S_1} + 4\lambda^{(A)}_{S_2}) \\
&+ \mathrm{lambdaT3GeneralQ000211I1}(4\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big) + 4(4\,(\mathrm{lambdaT3GeneralQ000211I1})^* + 5\,(\mathrm{lambdaT3Gener} \\
&+ 4\,(\mathrm{lambdaT3GeneralQ000211I1})^*\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^\dagger\Big) \\
&- \frac{22}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&- \frac{22}{15}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&+ 2\lambda_5\,(\lambda_5)^* \\
&+ 4\mathrm{lambdaT3GeneralQ001300I1}(5\,(\mathrm{lambdaT3GeneralQ000211I1})^* + (\mathrm{lambdaT3GeneralQ000211I2})^*) \\
&+ 4\mathrm{lambdaT3GeneralQ000211I2}((\mathrm{lambdaT3GeneralQ000211I2})^* + (\mathrm{lambdaT3GeneralQ001300I1})^*) \\
&+ 2\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^* \\
&+ \frac{63{g_2}^4}{5} \\
&- 24{g_2}^2\lambda^{(1)}_{12} \\
&- 6{g_1}^2\lambda^{(1)}_{12} \\
&+ 4{\lambda^{(1)}_{12}}^2 \\
&+ \lambda^{(1)}_{S_1}(8\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}) \\
&+ \lambda^{(1)}_{S_2}(8\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}) \\
&+ 16{\lambda^{(A)}_{12}}^2 \\
&+ 4{\lambda^{(\times)}_{12}}^2 \\
&+ 4\lambda^{(\times)}_{12}\lambda^{(A)}_{S_1} \\
&+ 4\lambda^{(\times)}_{12}\lambda^{(A)}_{S_2} \\
&+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_5}{d\ln\mu} = {}& \lambda_5\left(T^{(1)}_\nu + \mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + T^{(2)}_\nu + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) - \frac{4\lambda^{(A)}_{12}}{3} - 2\lambda_{12}\right. \\
&+ 2\,(\mathrm{lambdaT3GeneralQ021010I1})^*\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big) \\
&+ 2\,(\mathrm{lambdaT3GeneralQ021010I1})^*\,(2\,(\mathrm{lambdaT3GeneralQ000211I1})^* - (\mathrm{lambdaT3GeneralQ000211I2})^* + 4\,(\mathrm{la}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{33{g_2}^2\lambda_5}{2}\\
&- \frac{9{g_1}^2\lambda_5}{2}\\
&+ 2\lambda_{12}^{(1)}\lambda_5\\
&+ 2\lambda_1\lambda_5\\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5\\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} &= \lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_1}^{(A)})\\
&- \frac{4}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&+ \frac{4}{3}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- \frac{4}{3}\lambda_5\,(\lambda_5)^*\\
&- \frac{8}{9}\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^*\\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}\\
&- \frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}\\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}\\
&+ 2\lambda_1\lambda_{HS_1}^{(A)}\\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}\\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} &= \lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)})\\
&+ \frac{4}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- \frac{8}{9}\lambda_5\,(\lambda_5)^*\\
&- \frac{8}{9}\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^*\\
&+ {g_1}^2\left(4{g_2}^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right)\\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}\\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}\\
&+ 2\lambda_1\lambda_{HS_2}^{(A)}\\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}\\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda^{(A)}_{S_1}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda^{(A)}_{S_1}}{d\ln\mu} = {}& \lambda^{(A)}_{S_1}(4T^{(1)}_{\nu} + 4\,\mathrm{Tr}\Big(\text{y1Conj}\ (\text{y1Conj})^{\dagger}\Big)) \\
&+ \text{lambdaT3GeneralQ001300I1}\left(-\frac{4}{3}\,\mathrm{Tr}\Big(y_1\ (\text{y1Conj})^{\dagger}\Big) - \frac{44\,(\text{lambdaT3GeneralQ001300I1})^{*}}{3}\right) \\
&- \frac{4}{3}\,(\text{lambdaT3GeneralQ001300I1})^{*}\,\mathrm{Tr}\Big(\text{y1Conj}\ (y_1)^{\dagger}\Big) \\
&+ \frac{76}{45}\,\mathrm{Tr}\Big(y_1\ (y_1)^{\dagger}\ \text{y1Conj}\ (\text{y1Conj})^{\dagger}\Big) \\
&+ \frac{76}{45}\,\mathrm{Tr}\Big(y_1\ (\text{y1Conj})^{\dagger}\ \text{y1Conj}\ (y_1)^{\dagger}\Big) \\
&- \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_1\ (y_1)^{\dagger}\Big) \\
&- \frac{2}{3}\,\mathrm{Tr}\Big(\text{y1Conj}\ (\text{y1Conj})^{\dagger}\ \text{y1Conj}\ (\text{y1Conj})^{\dagger}\Big) \\
&- \frac{4}{3}\text{lambdaT3GeneralQ000211I2}\,(\text{lambdaT3GeneralQ000211I1})^{*} \\
&- \frac{4}{3}\text{lambdaT3GeneralQ000211I1}(3\,(\text{lambdaT3GeneralQ000211I1})^{*} + (\text{lambdaT3GeneralQ000211I2})^{*}) \\
&- \frac{224}{3}\text{lambdaT3GeneralQ000400I1}\,(\text{lambdaT3GeneralQ000400I1})^{*} \\
&- \frac{8}{3}\text{lambdaT3GeneralQ110200I1}\,(\text{lambdaT3GeneralQ110200I1})^{*} \\
&- 2{g_2}^4 \\
&- 24{g_2}^2\lambda^{(A)}_{S_1} \\
&+ 12{\lambda^{(A)}_{12}}^2 \\
&- \frac{4{\lambda^{(\times)}_{12}}^2}{3} \\
&+ 3{\lambda^{(A)}_{HS_1}}^2 \\
&+ 12\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1} \\
&+ 20{\lambda^{(A)}_{S_1}}^2
\end{aligned}
$$

$\lambda^{(A)}_{S_2}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda^{(A)}_{S_2}}{d\ln\mu} = {}& 4\lambda^{(A)}_{S_2}T^{(2)}_{\nu} \\
&- \frac{2}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big) \\
&- \frac{4}{3}\text{lambdaT3GeneralQ000211I2}\,(\text{lambdaT3GeneralQ000211I2})^{*} \\
&- 2{g_2}^4 \\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda^{(A)}_{S_2}\right) \\
&- 24{g_2}^2\lambda^{(A)}_{S_2} \\
&+ 12{\lambda^{(A)}_{12}}^2 \\
&- \frac{4{\lambda^{(\times)}_{12}}^2}{3}
\end{aligned}
$$

$$
\begin{aligned}
&+ 3{\lambda_{HS_2}^{(A)}}^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{12}^{(A)}}{d\ln\mu} = {}& \lambda_{12}^{(A)}\left(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \left(\mathrm{y1Conj}\right)^\dagger\Big) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3}\right) \\
&+ \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ \left(y_1\right)^\dagger\ y_2\ \left(y_2\right)^\dagger\Big) \\
&- \frac{2}{3}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \left(\mathrm{y1Conj}\right)^\dagger\ y_2\ \left(y_2\right)^\dagger\Big) \\
&- \frac{2}{9}\lambda_5\left(\lambda_5\right)^* \\
&- \frac{1}{3}\mathrm{lambdaT3GeneralQ021010I1}\left(\mathrm{lambdaT3GeneralQ021010I1}\right)^* \\
&- 24{g_2}^2\lambda_{12}^{(A)} \\
&- 6{g_1}^2\lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$\lambda_{12}^{(\times)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{12}^{(\times)}}{d\ln\mu} = {}& \lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \left(\mathrm{y1Conj}\right)^\dagger\Big) + 2T_\nu^{(2)}) \\
&+ \mathrm{lambdaT3GeneralQ000211I2}(2\,\mathrm{Tr}\Big(y_1\ \left(\mathrm{y1Conj}\right)^\dagger\Big) + 8\left(\mathrm{lambdaT3GeneralQ000211I1}\right)^* + 10\,(\mathrm{lambdaT3Genera} \\
&+ 2\left(\mathrm{lambdaT3GeneralQ000211I2}\right)^*\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \left(y_1\right)^\dagger\Big) \\
&- \frac{4}{5}\,\mathrm{Tr}\Big(y_1\ \left(y_1\right)^\dagger\ y_2\ \left(y_2\right)^\dagger\Big) \\
&- \frac{4}{5}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \left(\mathrm{y1Conj}\right)^\dagger\ y_2\ \left(y_2\right)^\dagger\Big) \\
&- \lambda_5\left(\lambda_5\right)^* \\
&+ 8\mathrm{lambdaT3GeneralQ000211I1}\left(\mathrm{lambdaT3GeneralQ000211I2}\right)^* \\
&+ 4\mathrm{lambdaT3GeneralQ001300I1}\left(\mathrm{lambdaT3GeneralQ000211I2}\right)^* \\
&- \mathrm{lambdaT3GeneralQ021010I1}\left(\mathrm{lambdaT3GeneralQ021010I1}\right)^* \\
&+ \frac{51{g_2}^4}{10} \\
&- 24{g_2}^2\lambda_{12}^{(\times)} \\
&- 6{g_1}^2\lambda_{12}^{(\times)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ 10{\lambda_{12}^{(\times)}}^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}\\
&- 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}\\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}\\
&- 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}
\end{aligned}
$$

$y1Conj$

$$
\begin{aligned}
16\pi^2\,\frac{dy1Conj}{d\ln\mu} &= 2\,\mathrm{Tr}\left(\mathrm{y1Conj}\ \left(y_1\right)^\dagger\right)\left(y_1\right)_{ij}\\
&+ T_\nu^{(1)}\left(\mathrm{y1Conj}\right)_{ij}\\
&+ \mathrm{Tr}\left(\mathrm{y1Conj}\ \left(\mathrm{y1Conj}\right)^\dagger\right)\left(\mathrm{y1Conj}\right)_{ij}\\
&+ \frac{1}{12}(29\left(y_1\ \left(y_1\right)^\dagger\ \mathrm{y1Conj}\right)_{ij} + 29\left(\mathrm{y1Conj}\ \left(y_1\right)^\dagger\ y_1\right)_{ij} + 9\left(y_2\ \left(y_2\right)^\dagger\ \mathrm{y1Conj}\right)_{ij} + 6\left(Y_e\ \left(Y_e\right)^\dagger\ \mathrm{y1Conj}\right)_{ij} +\\
&- \frac{9}{2}{g_2}^2\left(\mathrm{y1Conj}\right)_{ij}\\
&- \frac{3}{2}{g_1}^2\left(\mathrm{y1Conj}\right)_{ij}
\end{aligned}
$$

$lambdaT3GeneralQ000211I1$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ000211I1}{d\ln\mu} &= \lambda_{12}^{(1)}(2\,\mathrm{Tr}\left(\mathrm{y1Conj}\ \left(y_1\right)^\dagger\right) + 8\mathrm{lambdaT3GeneralQ000211I1} + 10\mathrm{lambdaT3GeneralQ001}\\
&+ \mathrm{lambdaT3GeneralQ000211I1}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(\mathrm{y1Conj}\ \left(\mathrm{y1Conj}\right)^\dagger\right) + 2T_\nu^{(2)})\\
&- \frac{22}{15}\,\mathrm{Tr}\left(\mathrm{y1Conj}\ \left(y_1\right)^\dagger\ y_2\ \left(y_2\right)^\dagger\right)\\
&+ 2\left(\lambda_5\right)^*\left(\mathrm{lambdaT3GeneralQ021010I1}\right)^*\\
&+ 8\mathrm{lambdaT3GeneralQ000400I1}(5\left(\mathrm{lambdaT3GeneralQ000211I1}\right)^* + (\mathrm{lambdaT3Gene}\\
&- 24{g_2}^2\mathrm{lambdaT3GeneralQ000211I1}\\
&- 6{g_1}^2\mathrm{lambdaT3GeneralQ000211I1}\\
&+ \lambda_{12}^{(\times)}(4\mathrm{lambdaT3GeneralQ000211I2} + 4\mathrm{lambdaT3GeneralQ001300I1})\\
&+ 4\lambda_{HS_2}^{(1)}\mathrm{lambdaT3GeneralQ110200I1}\\
&+ 2\lambda_{S_1}^{(1)}\mathrm{lambdaT3GeneralQ000211I1}\\
&+ \lambda_{S_1}^{(A)}(-4\mathrm{lambdaT3GeneralQ000211I1} - 2\mathrm{lambdaT3GeneralQ000211I2})\\
&+ \lambda_{S_2}^{(1)}(8\mathrm{lambdaT3GeneralQ000211I1} + 2\mathrm{lambdaT3GeneralQ000211I2})\\
&+ \lambda_{S_2}^{(A)}(4\mathrm{lambdaT3GeneralQ000211I1} + 2\mathrm{lambdaT3GeneralQ000211I2})
\end{aligned}
$$

$lambdaT3GeneralQ000211I2$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ000211I2}{d\ln\mu} &= \lambda_{12}^{(\times)}(4\,\mathrm{Tr}\left(\mathrm{y1Conj}\ \left(y_1\right)^\dagger\right) + 16\mathrm{lambdaT3GeneralQ000211I1} + 20\mathrm{lambdaT3GeneralQ0}\\
&+ \mathrm{lambdaT3GeneralQ000211I2}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\left(\mathrm{y1Conj}\ \left(\mathrm{y1Conj}\right)^\dagger\right) + 2T_\nu^{(2)})\\
&- \frac{8}{5}\,\mathrm{Tr}\left(\mathrm{y1Conj}\ \left(y_1\right)^\dagger\ y_2\ \left(y_2\right)^\dagger\right)\\
&- 2\left(\lambda_5\right)^*\left(\mathrm{lambdaT3GeneralQ021010I1}\right)^*\\
&+ 16\mathrm{lambdaT3GeneralQ000400I1}\left(\mathrm{lambdaT3GeneralQ000211I2}\right)^*
\end{aligned}
$$

$$
\begin{aligned}
&- 24{g_2}^2\text{lambdaT3GeneralQ000211I2}\\
&- 6{g_1}^2\text{lambdaT3GeneralQ000211I2}\\
&+ 8\lambda_{12}^{(1)}\text{lambdaT3GeneralQ000211I2}\\
&+ 2\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ000211I2}\\
&+ 2\lambda_{S_1}^{(A)}\text{lambdaT3GeneralQ000211I2}\\
&+ 2\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ000211I2}\\
&- 2\lambda_{S_2}^{(A)}\text{lambdaT3GeneralQ000211I2}
\end{aligned}
$$

lambdaT3GeneralQ000400I1

$$
\begin{aligned}
16\pi^2\,\frac{d lambdaT3GeneralQ000400I1}{d\ln\mu} &= \text{lambdaT3GeneralQ000400I1}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(\text{y1Conj }(\text{y1Conj})^\dagger\Big))\\
&+ 2\text{lambdaT3GeneralQ001300I1}\,\mathrm{Tr}\Big(\text{y1Conj }(y_1)^\dagger\Big)\\
&- \frac{23}{15}\,\mathrm{Tr}\Big(\text{y1Conj }(y_1)^\dagger\ \text{y1Conj }(y_1)^\dagger\Big)\\
&- 24{g_2}^2\text{lambdaT3GeneralQ000400I1}\\
&+ 12\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ000400I1}\\
&- 8\lambda_{S_1}^{(A)}\text{lambdaT3GeneralQ000400I1}\\
&+ 3\text{lambdaT3GeneralQ000211I1}^2\\
&+ 2\text{lambdaT3GeneralQ000211I1lambdaT3GeneralQ000211I2}\\
&+ \text{lambdaT3GeneralQ000211I2}^2\\
&+ 11\text{lambdaT3GeneralQ001300I1}^2\\
&+ 2\text{lambdaT3GeneralQ110200I1}^2
\end{aligned}
$$

lambdaT3GeneralQ001300I1

$$
\begin{aligned}
16\pi^2\,\frac{d lambdaT3GeneralQ001300I1}{d\ln\mu} &= \lambda_{S_1}^{(1)}(2\,\mathrm{Tr}\Big(\text{y1Conj }(y_1)^\dagger\Big) + 22\text{lambdaT3GeneralQ001300I1})\\
&+ \text{lambdaT3GeneralQ000400I1}(8\,\mathrm{Tr}\Big(y_1\ (\text{y1Conj})^\dagger\Big) + 88\,(\text{lambdaT3GeneralQ001300I}\\
&+ \text{lambdaT3GeneralQ001300I1}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(\text{y1Conj }(\text{y1Conj})^\dagger\Big))\\
&- \frac{46}{15}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ \text{y1Conj }(y_1)^\dagger\Big)\\
&- \frac{46}{15}\,\mathrm{Tr}\Big(\text{y1Conj }(y_1)^\dagger\ \text{y1Conj }(\text{y1Conj})^\dagger\Big)\\
&- 24{g_2}^2\text{lambdaT3GeneralQ001300I1}\\
&+ \lambda_{12}^{(1)}(6\text{lambdaT3GeneralQ000211I1} + 2\text{lambdaT3GeneralQ000211I2})\\
&+ \lambda_{12}^{(\times)}(4\text{lambdaT3GeneralQ000211I1} + 4\text{lambdaT3GeneralQ000211I2})\\
&+ 4\lambda_{HS_1}^{(1)}\,\text{lambdaT3GeneralQ110200I1}
\end{aligned}
$$

lambdaT3GeneralQ021010I1

$$
\begin{aligned}
16\pi^2\,\frac{d lambdaT3GeneralQ021010I1}{d\ln\mu} &= 2\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_1\ (\text{y1Conj})^\dagger\Big)\\
&+ \text{lambdaT3GeneralQ021010I1}(T_\nu^{(1)} + \mathrm{Tr}\Big(\text{y1Conj }(\text{y1Conj})^\dagger\Big) + T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\,(\lambda_5)^{*}\,(2\,(\mathrm{lambdaT3GeneralQ000211I1})^{*} - (\mathrm{lambdaT3GeneralQ000211I2})^{*} + 4\,(\mathrm{la}\\
&- \frac{33{g_2}^{2}\mathrm{lambdaT3GeneralQ021010I1}}{2}\\
&- \frac{9{g_1}^{2}\mathrm{lambdaT3GeneralQ021010I1}}{2}\\
&+ 2\lambda_{12}^{(1)}\mathrm{lambdaT3GeneralQ021010I1}\\
&- 2\lambda_{12}^{(A)}\mathrm{lambdaT3GeneralQ021010I1}\\
&- 2\lambda_{12}^{(\times)}\mathrm{lambdaT3GeneralQ021010I1}\\
&+ 2\lambda_{1}\mathrm{lambdaT3GeneralQ021010I1}\\
&+ 4\lambda_{HS_1}^{(1)}\mathrm{lambdaT3GeneralQ021010I1}\\
&- \frac{4\lambda_{HS_1}^{(A)}\mathrm{lambdaT3GeneralQ021010I1}}{3}\\
&+ 4\lambda_{HS_2}^{(1)}\mathrm{lambdaT3GeneralQ021010I1}\\
&- \frac{4\lambda_{HS_2}^{(A)}\mathrm{lambdaT3GeneralQ021010I1}}{3}
\end{aligned}
$$

$lambdaT3GeneralQ110200I1$

$$
\begin{aligned}
16\pi^{2}\,\frac{dlambdaT3GeneralQ110200I1}{d\ln\mu} &= \lambda_{HS_1}^{(1)}(2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \,(y_1)^{\dagger}\Big) + 10\mathrm{lambdaT3GeneralQ001300I1} + 8\mathrm{lambdaT3GeneralQ11}\\
&+ \mathrm{lambdaT3GeneralQ110200I1}(2T_{\nu}^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \,(\mathrm{y1Conj})^{\dagger}\Big) + 6\,\mathrm{Tr}\Big(Y_d\ \,(Y_d)^{\dagger}\Big) +\\
&- 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \,(y_1)^{\dagger}\ Y_e\ \,(Y_e)^{\dagger}\Big)\\
&+ 4\,(\lambda_5)^{*}\,(\mathrm{lambdaT3GeneralQ021010I1})^{*}\\
&+ 40\mathrm{lambdaT3GeneralQ000400I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^{*}\\
&- \frac{33{g_2}^{2}\mathrm{lambdaT3GeneralQ110200I1}}{2}\\
&- \frac{3{g_1}^{2}\mathrm{lambdaT3GeneralQ110200I1}}{2}\\
&+ 6\lambda_{1}\mathrm{lambdaT3GeneralQ110200I1}\\
&+ \lambda_{HS_2}^{(1)}(6\mathrm{lambdaT3GeneralQ000211I1} + 2\mathrm{lambdaT3GeneralQ000211I2})\\
&+ 2\lambda_{S_1}^{(1)}\mathrm{lambdaT3GeneralQ110200I1}\\
&- 4\lambda_{S_1}^{(A)}\mathrm{lambdaT3GeneralQ110200I1}
\end{aligned}
$$

$muT3GeneralS1Sq$

$$
\begin{aligned}
16\pi^{2}\,\frac{dmuT3GeneralS1Sq}{d\ln\mu} &= m_1^2(2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \,(y_1)^{\dagger}\Big) + 10\mathrm{lambdaT3GeneralQ001300I1})\\
&- 4\,\mathrm{Tr}\Big(M_F\ \,(M_F)^{\dagger}\ \,(\mathrm{y1Conj})^{T}\ \,(y_1)^{*}\Big)\\
&+ \mathrm{muT3GeneralS1Sq}(2T_{\nu}^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ \,(\mathrm{y1Conj})^{\dagger}\Big))\\
&+ 40\mathrm{lambdaT3GeneralQ000400I1}\,(\mathrm{muT3GeneralS1Sq})^{*}\\
&- 12{g_2}^{2}\mathrm{muT3GeneralS1Sq}\\
&+ 2\lambda_{S_1}^{(1)}\mathrm{muT3GeneralS1Sq}\\
&- 4\lambda_{S_1}^{(A)}\mathrm{muT3GeneralS1Sq}\\
&+ m_2^2(6\mathrm{lambdaT3GeneralQ000211I1} + 2\mathrm{lambdaT3GeneralQ000211I2})
\end{aligned}
$$